Task 2
Marking Scheme

Chemistry, Physics, Biology – Tied by Time and Change

EOES2025, Zagreb, Croatia
26.04. – 03.05.2025.

Problem 1. Kinetics of the reaction between crystal violet and hydroxide ions

(TOTAL MARKS: 41.5)

-50% on the mark if units are missing or the number of decimals or notation of the final result (e.g. scientific notation) is not as required in the certain part of the task
-75% in total on the mark if both errors present

Step 1.1. Assembling the photometer (2 p)

To be circled and signed by a lab assistant after you raise the red card:	Marks
(a) The photometer is correctly assembled. ___*lab. assistant's signature*_ (2 pt) **(b) The photometer is not correctly assembled. The lab assistant provided the solution.** ___*lab. assistant's signature*_____ (0 pt) If the error is about assembling Lego bricks (non-scientific problem) – if it is not something completely wrong, the assistant will help, and 2 pt will be given If the problem is about the electric circuit (scientific problem: short circuit, resistor not connected, opposite polarities connected, and others), the assistant will connect everything properly, but no points will be given	**2**

Step 1.2: Preparation of solutions for measurement

1.2.1. Preparation of sodium hydroxide solutions

- *1.2.1. Fill in the Table 1.2.1 and detail your calculations only for the first line on the Answer Sheet below Table 1.2.1.! Indicate your final result for [NaCl] using the scientific notation with 3 significant figures (example: 1.21×10^{-5}). For calculated volumes of solutions A and B, indicate the result as a number with 2 significant figures.* (4.5 p)
-

Table 1.2.1. Data for preparation of solutions **I-III**

SOLUTION/ EXPERIMENT LABEL	[NaOH] (M)	[NaOH] + [NaCl] (M)	[NaCl] (M)	V(**A**) (mL)	V(**B**) (mL)	Marks
						4.5

I	1.20×10^{-2}	3.00×10^{-2}	1.80×10^{-2}	7.5	3.0	
II	9.60×10^{-3}	3.00×10^{-2}	2.04×10^{-2}	6.0	3.4	
III	7.20×10^{-3}	3.00×10^{-2}	2.28×10^{-2}	4.5	3.8	

0.5 pt for each correct value

Detailed calculation for the first line (1.5 p):

	Marks
[NaCl] = $(3.0\times10^{-2} - 1.2\times10^{-2})$ M = 1.8×10^{-2} M $V(\mathbf{A}) = V_{ST}(NaOH) = (c_2/c_1) \times V_2 = (1.2\times10^{-2}$ M / 0.08 M$) \times 50$ mL = 7.5 mL $V(\mathbf{B}) = V_{ST}(NaCl) = (c_2/c_1) \times V_2 = (1.8\times10^{-2}$ M / 0.30 M$) \times 50$ mL = 3.0 mL The values are already marked in the table above. Correct details of the calculation = 0.5 pts / each	**1.5**

1.2.2. Preparation of CV solution

- *1.2.2. Show details of your calculations. Indicate your final result for $V_{ST}(CV)$ as a number without decimal places in the μL units.* (1 p)

Detailed calculation:

	Marks
$V_{ST}(CV) = (c_2/c_1) \times V_2 = (2.16\times10^{-5}$ M / 4.80×10^{-4} M$) \times 10$ mL = 450 µL 0.5 pt details + 0.5 pt value	**1**

Step 1.3. Measurement of the reaction rate

1.3.1. Voltage measurement for the blank sample (distilled water) (1.6 p)

Checkpoint! To be filled and signed by a lab assistant after you raise the red card: **Voltage for the blank sample (water) = __**[about 1300]**_ mV** **LED diodes changed: Yes No**	*NO POINTS*

______________________________	

Table 1.3.1. Measured values of U_{BL}	Marks
	0.6

Measurement number	**U_{BL} (mV)**
1	
2	
3	

Express the measurements in mV units without decimals.

Each value 0.2 pt, all three values present 0.6 pt

- *1.3.1. Calculate the average value ($\overline{U_{BL}}$) for the blank sample. Show details of your calculations and indicate your final result for $\overline{U_{BL}}$ as a number without decimal places in mV units.*

Marks **1**

$$U_{\mathrm{BL}} = \frac{U_1 + U_2 + U_3}{3}$$

1.3.2. Temperature in the laboratory

Before continuing, raise the red card for the lab assistant to record the temperature in the lab. The rate constant is temperature-dependent and it is very important to record the temperature.

Temperature in the lab = _________ °C **Time: _______________** **Lab assistant's signature: __________________________________**	***NO POINTS***

Step 1.4. Data collecting and processing

1.4.1. Calculation of the absorbances and $ln(A_t)$ (11.8 p)

- *Fill in the **Tables 1.4.1.** with the measured voltages in mV (without decimal places). Absorbances and ln(A_t) must be indicated as numbers with 4 decimal places.*
- *Detail your calculations **only for the first line** of **Table 1.4.1**. for Experiment **I** only.*

Marks
10.8

Tables 1.4.1. Measured and calculated values.

Experiment **I**				
Data point	*t* (min)	*U* (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				
5.				
6.				
7.				
8.				

Experiment **II**				
Data point	*t* (min)	*U* (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				
5.				
6.				
7.				
8.				

Experiment **III**				
Data point	*t* (min)	*U* (mV)	A_t	$\ln(A_t)$
1.	0:00			
2.				
3.				
4.				
5.				
6.				

7.				
8.				

- For each pair of measured values (*U* at different *t*) 0.25 pt (total 6.0 pt)
- For each calculated value of *A* 0.1 pt (total 2.4 pt)
- For each calculated value of ln(*A*) 0.1 pt (total 2.4 pt)

- *Detail your calculation for the first line of Experiment* ***I****. Absorbances and ln(A) must be indicated as numbers with 4 decimal places.*

	Marks
$A_{CV} = log\left(\frac{U_{BL}}{U_{CV}}\right)$ $ln(A_{t=0}) = \cdots$ Value is already marked in the table, 0 pt in this part Details of calculation 1 pt	**1**

1.4.2. Calculation of the *k*' (11.1 p)

- *1.4.2.1. Draw a graph of* $\ln(A_t)$ *versus t on the millimeter paper. Data points must be shown as ×, points used for the calculation of the slope must be shown as ⊗. Label the paper with your team/country's sticker.*
- *1.4.2.2. Determine the reaction rate constants k' for each of the three measurements from* ***Graph 1.4.2.1.*** *Detail your calculations* ***only for the Experiment I****.* Express the values in units min^{-1} having 4 decimal places *and fill in the Table 1.4.2.*

	Marks
$k' = \frac{\ln(A_{t1}) - \ln(A_{t1})}{t_1 - t_2}$ Graph: 0.5 pt: labelling of axes (e. g. ln *A* / t (min)) 3×0.7 pt: regression lines (placement of the regression line on the graph (eye best fit), not taking into account individual points that deviate significantly 3×0.8 pt: precision of the graphs (spacing on the axes (1 pt) and positioning of the points (within 0.5 mm of table value; -0.1 pt for each error) 3×0.5 pt: marked points whose coordinates were taken for calculating the slope	**7.5**

1 pt: details of slope (k') calculation Calculated values of k' are not marked at this point to avoid a double penalty since the final value of k is marked in the last part, which reflects the accuracy of k' values (even if only two k' are present).	

Table 1.4.2. Determined values of k'

Marks

3.6

SOLUTION/ EXPERIMENT LABEL	k' (min^{-1})
I	
II	
III	

Temperature / °C	k' (I) / min^{-1}	k' (II) / min^{-1}	k' (III) / min^{-1}
21	0.0403	0.0322	0.0242
22	0.0438	0.0350	0.0263
23	0.0477	0.0382	0.0286
24	0.0518	0.0415	0.0311
25	0.0563	0.0451	0.0338
26	0.0612	0.0490	0.0367
27	0.0666	0.0533	0.0400
28	0.0720	0.0576	0.0432

Error of k' *(compared with known* k' *at the temperature measured in the lab)*	*Points for each of the three calculated* k'
<20%	1.2
21-39%	0.72
40-59%	0.36
60-79%	0.12
>80%	0

Error of k' is calculated using eqn:

$$error = \frac{k'(determined) - k'(known)}{k'(known)}$$	

1.4.3. Calculation of the rate constant *k* (8 p)

Before continuing, raise the red card for the lab assistant to record the temperature in the lab. The rate constant is temperature-dependent and it is very important to record the temperature.

Temperature in the lab = ________ °C **Time: _______________** **Lab assistant's signature: __________________________________**	***NO POINTS***

- *1.4.3.1. Assuming the OH^- concentration remains constant during the experiment, calculate its concentration in the cuvette for all three experiments. Results must be given using the scientific notation with 1 decimal place (example: $1.2×10^{-5}$).*

	Marks
Experiment **I**: $[OH^-] = 1.2×10^{-2} / 2 = 6.0×10^{-3}$ M Experiment **II**: $[OH^-] = 9.6×10^{-3} / 2 = 4.8×10^{-3}$ M Experiment **III**: $[OH^-] = 7.2×10^{-3} / 2 = 3.6×10^{-3}$ M - Each correct value 0.5 pt (total 1.5 pt)	**1.5**

- *1.4.3.2. Draw the graph on the millimeter paper using the data from **Table 1.4.2.** and concentrations of $[OH^-]$ in all three experiments from part 1.4.3.1. Elements on the graph that must be shown: axes with titles and values, <u>all</u> k' – $[OH^-]$ values, regression lines, data points must be shown as ×, points used for the calculation of the slope must be shown as ⊗. Label the paper with your team/country's sticker.*

- *1.4.3.3. From **Graph 1.4.3.1.** calculate the value of the rate constant k. Show the details of your calculations. Final value of k that you report must be in M^{-1} min^{-1} units and have 2 significant figures.*

Detailed calculation of *k*:

<table>
<tr><td>
Graph:

0.5 pt: labelling of axes (e. g. k (min^{-1}) / [OH^-] (M))

0.7 pt: regression line (placement of the regression line on the graph (eye best fit), not taking into account individual points that deviate significantly

0.3 pts: precision of the graph (spacing on the axes (1 pt) and positioning of the points (within 0.5 mm of true value; -0.1 pt for each error)

0.5 pt: marked points whose coordinates were taken for calculating the slope

1 pt: details of slope (k) calculation

$$k = \frac{k'_1 - k'_2}{[OH^-]_1 - [OH^-]_2}$$

k value marking: Temperature will be measured in each lab by the lab assistants.

It is common for chemical reactions that k depends on temperature. The empirical rule is that the reaction is 2x faster for every 10 °C increase in temperature (that is a 10% change in k for every 1 °C). For this specific reaction and conditions, we determined temperature dependence (k at different temperatures) which will be used in marking of the value of the k:
</td><td>Marks

6.5</td></tr>
</table>

Temperature / °C	*k* (known) / M^{-1} min^{-1}
21	6.71
22	7.30
23	7.95
24	8.64
25	9.39
26	10.2
27	11.1
28	12.0

Error of k (compared with known k at the temperature measured in the lab)	*Points for calculated k*
<20%	3.5
21-39%	2.1
40-59%	1.05
60-79%	0.35
>80%	0

Error of k is calculated using eqn: $$error = \frac{k(determined) - k(known)}{k(known)}$$	

Problem 2. Velocity and Drag

(TOTAL MARKS: 43.5)

<table>
<tr><td>2.1, 2.2, 2.4 and 2.5 (Tables on next page)</td><td>Marks</td></tr>
<tr><td>Every angle-vs-time table: 3.2 (0.4 per single entry)
Every pendulum period: 0.4
Bottom right table: every row 0.25

Note:
- values in tables may differ from student to student
- time values in angle-tables can be written with or without hundredth of a second
- pendulum period should be written to at least 3 decima digits (will be penalized by 0.1 p)
- pendulum periods should not decrease as dimension L increases (will be penalized by 0.4 p)</td><td>19</td></tr>
</table>

2.1 (3.2 p), *2.2* (0.4 p), *2.4* (14.4 p) *and 2.5* (1 p).

Paper width (cylinder height) L = 3.5 cm		
angle θ (deg)	**time t (s)**	**time t (s)**
20	0	0
13	**27.87**	**27.83**
8	**64.68**	**61.67**
5	**109.18**	**109.27**
3	**163.78**	**162.13**
Pendulum period: **1.532 s**		

Paper width (cylinder height) L = 5.0 cm		
angle θ (deg)	**time t (s)**	**time t (s)**
20	0	0
13	**23.15**	**23.15**
8	**53.99**	**54.02**
5	**87.80**	**89.21**
3	**103.25**	**126.07**
Pendulum period: **1.535 s**		

Paper width (cylinder height) L = 7.0 cm		
angle θ (deg)	**time t (s)**	**time t (s)**
20	0	0
13	**15.49**	**16.99**
8	**40.24**	**41.71**
5	**69.43**	**70.96**
3	**103.25**	**106.31**
Pendulum period: **1.536 s**		

Paper width (cylinder height) L = 14.0 cm		
angle θ (deg)	**time t (s)**	**time t (s)**
20	0	0
13	**7.83**	**9.46**
8	**21.77**	**23.3**
5	**38.86**	**41.99**
3	**65.21**	**66.65**
Pendulum period: **1.544 s**		

Paper width (cylinder height) L = 21.0 cm		
angle θ (deg)	**time t (s)**	**time t (s)**
20	0	0
13	**6.43**	**6.43**
8	**15.58**	**15.71**
5	**28.14**	**28.21**
3	**46.91**	**46.78**
Pendulum period: **1.558 s**		

angle θ (deg)	**logarithm of the angle**
20	3.00
13	**2.56**
8	**2.08**
5	**1.61**
3	**1.10**

Table 2.2. Measured values of the times for different paper cylinder dimension, at fixed angles 3 deg, 5deg, 8 deg and 13 deg. (20 deg corresponds to initial condition at t = 0.) Bottom right Table: logarithm of the angles.

2.3. Calculate the period of oscillation using equation (2.2). Determine the relative error between the measured and the calculated period values and enter them in the Answer Sheet. Show your calculations. (3 p)

		Marks
Value of R	$R =$____**58.0 cm**__	**1**
Calculated value of period T	$T =$___**1.528 s**____	**1**
Relative error between measured and calculated period T	relative error=____**0.26 %**___	**1**
Calculation		

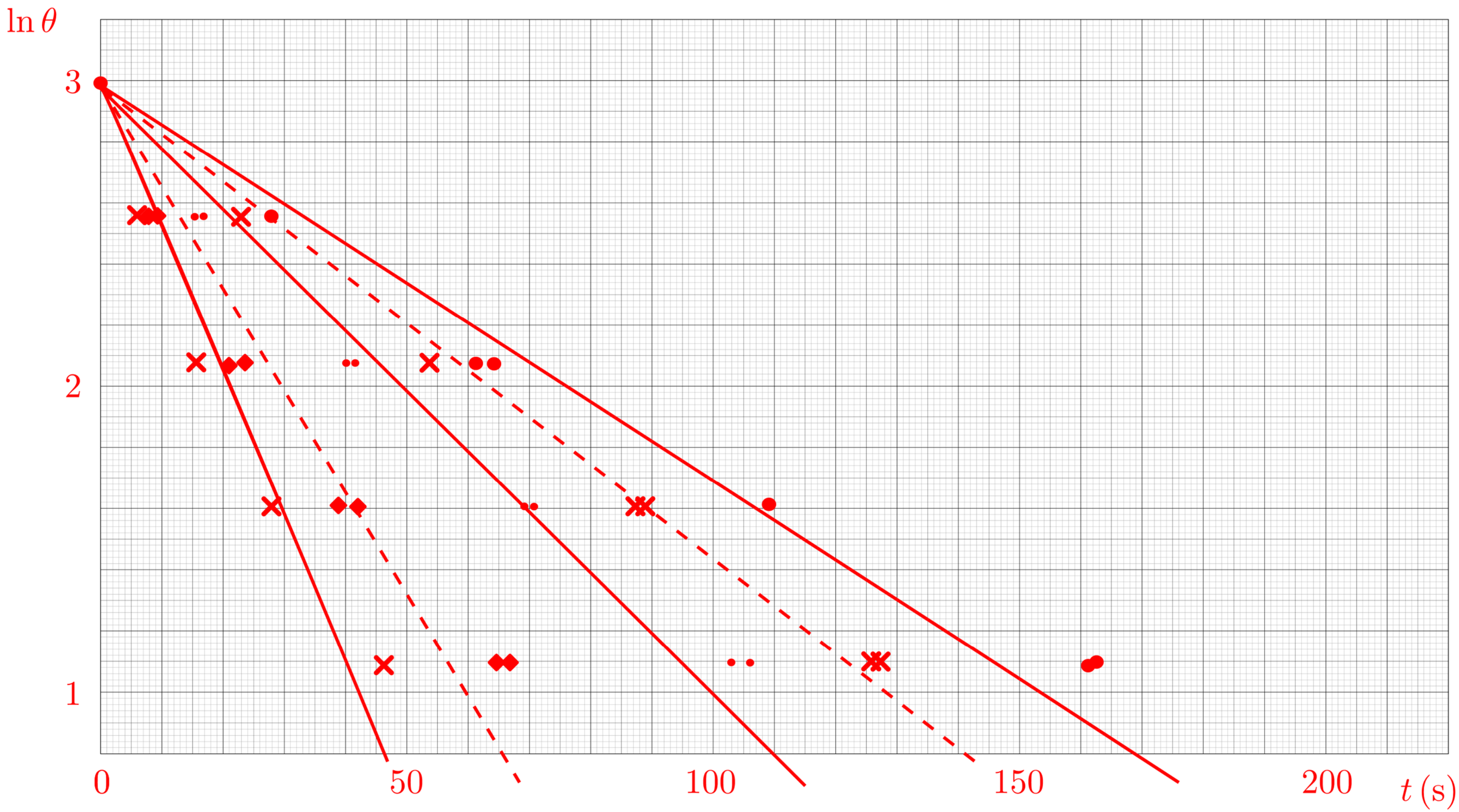
- string length to the top of the plastic body = 56.2 cm (may differ from student to student, by 1 cm)
- add distance to center of mass (1.8 cm)
- R = 56.2 + 1.8 = 58.0 cm
- T is calculated by Eq 2.2, and is equal to 1.528 s
- relative error = 0.26% or 0.0026 (may differ from student to student)

2.6. Given the measured values of the oscillation period T, is the drag caused by the paper cylinder negligible*, or* non-negligible*? Write the answer in the Answer Sheet. Show your calculation.* (1 p)

		Marks
The drag is	☒ ***negligible***	**0.5**
	☐ **non-*negligible***	
Period T	☐ ***decreases with height***	**0.5**
	☒ **increases with height**	
Calculation		

- max difference is between L = 2.5 cm and L = 21 cm cases, and is equal to 1.558 – 1.532 = 0.026 s
- 0.026/1.532 = 1.70% (or 0.026/1.558 = 1.67%)

2.7. Using the values from Table 2.2, plot time dependence of calculated values of the logarithm of the angle ($\ln\theta$), on graphing paper (Plot 2.1) in the Answer Sheet. Plot ALL data, without averaging. (6.5 p)



Plot 2.1. Graph of $\ln\theta$-vs-t at fixed angles, for different paper cylinder dimensions L.

	Marks
Labeling of axes (0.5 pt) **Data points for each length L: 0.5 (tot 2.5)**	**6.5**

Best fit lines: 0.7 (tot 3.5) (all values/points/lines may differ from student to student)	

2.8. For each length L, draw a straight line that best follows the points, and from the slope of that line determine the damping factor γ to three significant figures. Fill in the first two columns of Table 2.3. in the Answer Sheet with the obtained values. (7.5 p)

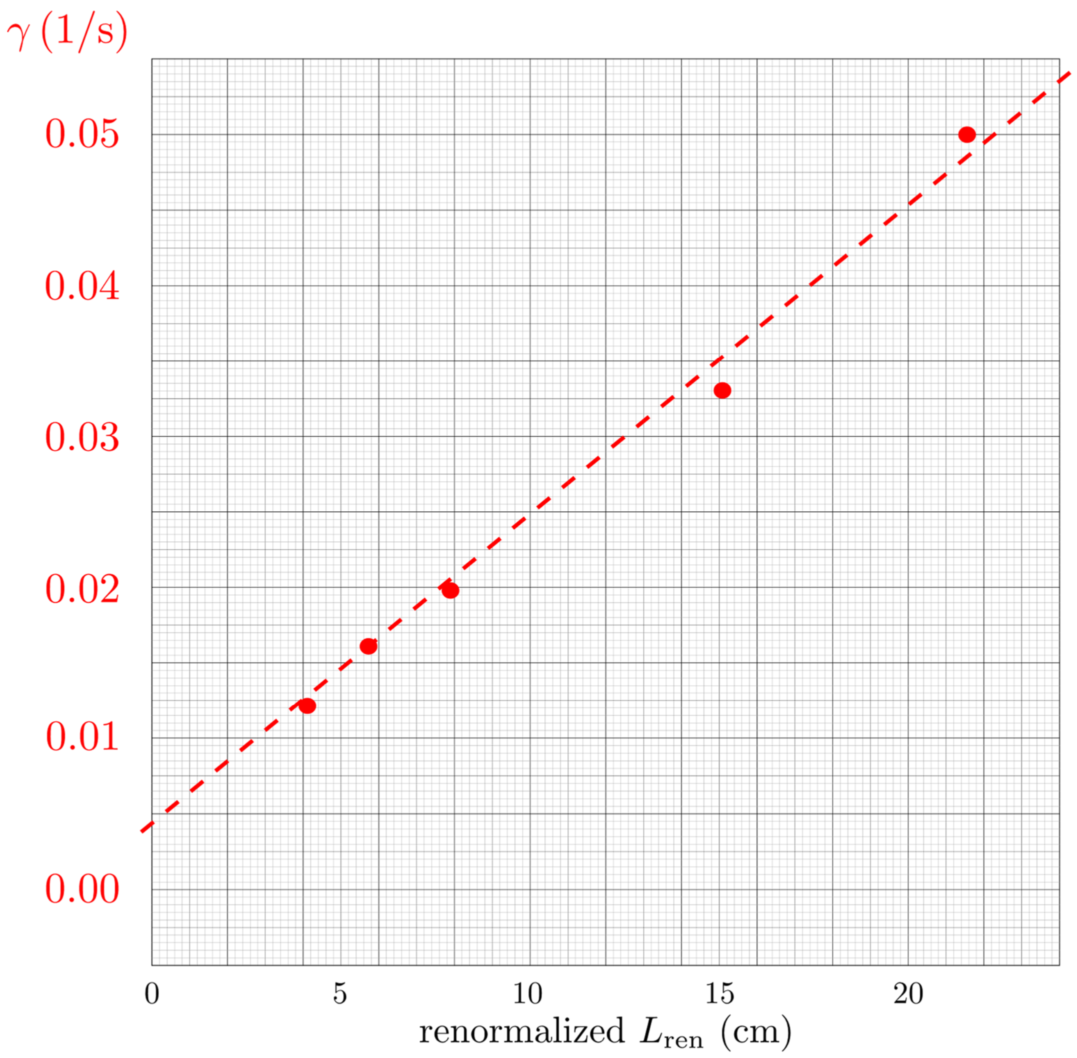
<table>
<tr><th>L (cm)</th><th>γ (1/s)</th><th>renormalized L_{ren} (cm)</th><th>Marks</th></tr>
<tr><td>3.5</td><td>0.0122</td><td>3.41</td><td rowspan="5">10

Every row 2 p (1.5 for γ, 0.5 for L_{ren})
All values may differ from student to student.
All values (except L) should have at least three significant digits.</td></tr>
<tr><td>5</td><td>0.0162</td><td>4.81</td></tr>
<tr><td>7</td><td>0.0198</td><td>6.63</td></tr>
<tr><td>14</td><td>0.0330</td><td>12.61</td></tr>
<tr><td>21</td><td>0.0507</td><td>18.01</td></tr>
</table>

Table 2.3.

2.9. For each value of the cylinder dimension L_i, determine the corresponding renormalized value, and enter the value in the last column of Table 2.3 in the Answer Sheet. (2.5 p)

2.10. On the graphing paper (Plot 2.2) in the Answer Sheet, plot values of γ as a function of the renormalized *dimension* L_{ren}. (2 p)



Plot 2.2. Dependence of γ on the renormalized dimension L_{ren}.

	Marks
Labeling of axes (0.5 pt) **Data points renormalized dimension L: 0.1 (tot 0.5)** **Best fit lines: 1** (all values/points may differ from student to student)	**2**

2.11. Using a ruler, draw the line of best fit on the graphing paper (Plot 2.2), and from its slope determine the value of constant C and enter it on the Answer Sheet. Show your calculations. (2 p)

		Marks
Value of constant C	C = **0.00672 kg/s m**	**2**
Calculation		
Line slope = (0.0535 – 0.0043)/0.22 = 0.224 1/s m (0.5 p) **C = 0.224 1/s m * 0.030 kg = 0.00672 kg/s m (1.5 p)** **Values in tables may differ from student to student.** **Obtained value must be within 35% of the value given here (i.e. between 0.00437 kg/s m and 0.00907 kg/s m).**		

Problem 3. Heat it up! Speed it up! Step it up!

(TOTAL MARKS: 40)

From Table 3.1.1. to 3.3.2. we have an Excel sheet that automatically shows correct numbers, once we enter data from Table 3.1.1.

STEP 3.1. Complete the Table 3.1.1. by entering the respective treatment temperatures (t_1 for the lower temperature and t_2 for the higher temperature), volumes of expelled water for each temperature treatment for each tube and mean volumes of expelled water for each temperature treatment (V1 for the mean at t_1, and V2 for the mean at t_2). If you have inconsistent volume, eliminate the tube with extreme minimum (outlier).

Table 3.1.1. *Treatment temperatures and volumes of expelled water.* (2 p)

	t_1	t_2
Temperature / °C		
Tube 1 ***Volume of expelled water / mL***		
Tube 2 ***Volume of expelled water / mL***		
Tube 3 ***Volume of expelled water / mL***		
Tube 4 ***Volume of expelled water / mL***		
Tube 5 ***Volume of expelled water / mL***		
Mean volume of expelled water / mL ***Round to one decimal place***	*X*	*X*
Marks	**1**	**1**

If all columns are filled with numbers, and the mean volume is calculated correctly (we will check in the Excel table) and round to one decimal place, students get 1 point for column t1 and one point for column t2.

STEP 3.2.

3.2.1. *Write down the Q_{10} coefficient that you obtained from the yeast fermentation experiment.* ***Round the number to one decimal place.*** (3 p)

The Q_{10} is:	Marks
	3
When you calculate the Q_{10}, please call the lab assistant for a check-up. If it is correct, carry on with your assignment. If wrong, the lab assistant will give you the right answer, but you will lose 3 points.	Lab assistant signature

If the student calculated Q10 coefficient obtained from the yeast fermentation experiment by himself/herself correctly, and it is rounded to one decimal place, he/she will get full points (3 points). We will check the calculation in our Excel sheet. If the student calculated wrongly, they get 0 points.

STEP 3.3.

Table 3.3.1. Write down the generation time of bacteria in YOUR rice bowl by using the Q_{10} coefficient obtained from the yeast experiment. In your case, you have a given generation time at 4 °C which is 8.0 h. **Round generation time to one decimal place and use this value for following calculations.** (1.5 p)

	Your results	Marks
Q_{10} coefficient		
Generation time (h) at 4°C	8.0	–
Generation time (h) at 14°C		**0.5**
Generation time (h) at 24°C		**0.5**
Generation time (h) at 34°C		**0.5**

Students need to divide each consequential generation time by Q_{10} coefficient, starting from 8.0. For each correct calculation they get 0.5 points. Number needs to be correctly round to one decimal place. It will be checked in Xcel sheet.

Table 3.3.2. *Complete the missing data in in the Table below. Determine the number of divisions (n) that occur over a 24-hour period at each temperature.* ***Round the number of divisions to the whole number.*** *Use the obtained data to calculate the final numbers of bacterial cells (N) at each temperature. Write N in scientific format.* (4 p)

Temperature (°C)	Generation time (h)	N_0	*n* – number of divisions in 24 h	*N*	Marks *(n)*	Marks *(N)*
4	8.0	500			**0.5**	**0.5**
14		500			**0.5**	**0.5**
24		500			**0.5**	**0.5**
34		500			**0.5**	**0.5**

The generation time in the Table 3.3.2. just needs to be rewritten from table 3.3.1. Number of divisions is calculated by dividing 24/generation time and round to a whole number. For each correctly calculated number of divisions students get 0.5 points. Final numbers (N) is calculated using the formula from Task sheet. If it is correct, and written in scientific format – X.xx × 10^x, in which case they get 0.5 points. If the number is wrong, or it is not written in scientific format, they get 0 points.

STEP 3.4.

Answer the following questions.

3.4.1. (1 p)

Question	***Answer***	Marks
What was the mass of glucose in your labelled package? (students must add unit)	**3 g**	**1**

If the answer is correct student will get full point (1.0 point), if number is wrong or the mass unit is missing, they get 0 points.

3.4.2. (1 p)

Question	***Answer***		Marks
Carefully look at the experimental setup from **Step 3.1.** and answer this: If the experiment had continued for another three hours, would the liquid have overflowed from the plastic syringe? Round the correct statement.	**YES**	**NO**	**1**

The correct answer is NO. This is because the needle in the experiment is too short, and at around 4.2 mL the needle is outside the water level. Students should be able to deduce this. Help will be that at 37°C, all the volumes of expelled water will be around 4.2 mL. This can give them a hint. If they just atke a lucky guess, they still get a point for correct answer.

3.4.3. **Question:** The experiment used expelled water as an indirect measure of yeast metabolism. Which of the following could have been another reliable way to measure the metabolic rate of yeast in the same experiment? **There can be one or more than one correct answers**, so mark it/them with an X. (1 p)

		Answer	Marks
A	Counting the number of yeast cells under a microscope before and after incubation.		
B	Measuring the amount of CO_2 gas produced during fermentation.	**X**	**1**
C	Measuring the amount of O_2 gas produced during fermentation.		
D	Measuring the amount of lactic acid produced during fermentation.		
E	Measuring the amount of propionic acid produced during fermentation.		

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at a B, and nothing else.

3.4.4. In the tubes, the yeast underwent fermentation. However, this included an essential biochemical process, which is: **There can be one or more than one correct answers**, so mark it/them with an X. (1 p)

		Answer	Marks
A	Glycolysis, which requires oxygen.		
B	Glycolysis, which does not require oxygen.	**X**	**1**
C	Citric acid cycle, which requires oxygen.		
D	Citric acid cycle, which does not require oxygen.		
E	Calvin cycle, which requires oxygen.		
F	Calvin cycle, which does not require oxygen.		

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at a B, and nothing else.

***3.4.5.* Question:** A bakery wants to optimize dough fermentation for faster rising. Based on your knowledge gained in this experiment, and your previous knowledge, which of the following would be the best advice for the bakers? **There may be one or more than one correct answers**, so mark it/them with an X. (1 p)

		Answer	Marks
A	Increase the dough temperature to 30 °C.	**X**	**1**
B	Increase the dough temperature to 60 °C.		
C	Add some artificial sweetener to the dough.		
D	Add some table sugar to the dough.	**X**	
E	Add a small amount of ethanol to pre-activate fermentation enzymes.		

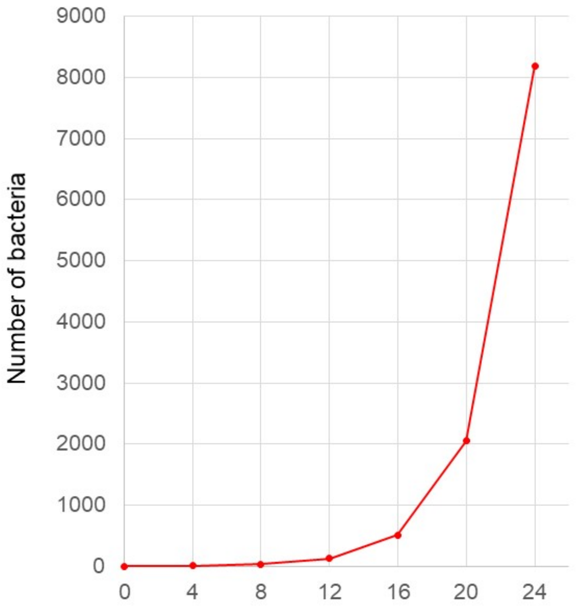
If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at both A and B, and nothing else. If only A, or B is marked this is not considered as correct answer, and students get 0 points.

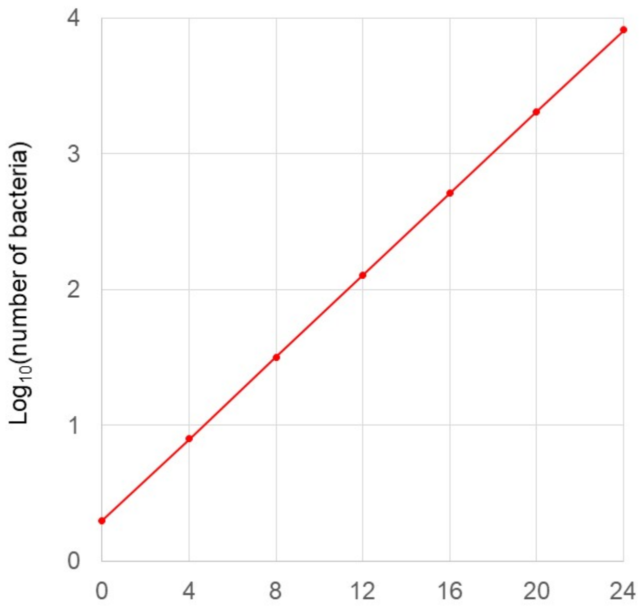
3.4.6.* Question:** Examine the results obtained in ***Table 3.3.2. - the number of bacteria after 24 h at different temperatures. Assume that anything under a thousand bacteria in the bowl is safe to consume. Now answer the following questions (1 p):

		Answer		Marks
A	If the starting number of bacteria was 1000, would the food be OK to eat after one day if kept at 34 °C?	**YES**	**NO**	**1**
B	If the starting number of bacteria was 10, would the food be OK to eat after one day if kept at 34°C?	**YES**	**NO**	
C	What if there was just a single bacterium in the bowl, would it be OK to eat after one day kept at 34°C?	**YES**	**NO**	
D	How about if there were no bacteria at all in the bowl, would it be OK to eat after one day kept at 34 °C?	**YES**	**NO**	

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Correct answer is only if they round everything like above.

3.4.7. *This will need some mathematical thinking, as well as artistic touch. Draw a curve of bacterial growth with following data:* ***generation time of 2 h****, and starting number of bacteria of 2 cells.* (2 p)





	Marks
Graph 1	**1**
Graph 2	**1**

If the drawings on each of the graphs are accurate student will get full point (1.0). The lines don't need to be perfect, but exponential growth line, and straight log line must be clearly and nicely drawn.

***3.4.8.* Question:** Undeniable evidence that food quality is highly dependent on storage temperature and the initial number of bacteria is all around you—even if you haven't noticed it until now! :)

In the following table, **all statements are factually correct**, but your task is to mark with an "X" only those that conclusively prove that an **increase in storage temperature and the initial bacterial presence** in food are key factors affecting its safety. (1 p)

		Answer	Marks
A	Cheese kept in the refrigerator will be ok for some time, but it will spoil much faster if kept at warm place.	**X**	**1**
B	Some types of milk can be kept in refrigerator, but will spoil pretty quickly if kept at warm place. On the other hand, some types of milk are kept in the supermarket outside the refrigerator, but still they do not spoil.	**X**	

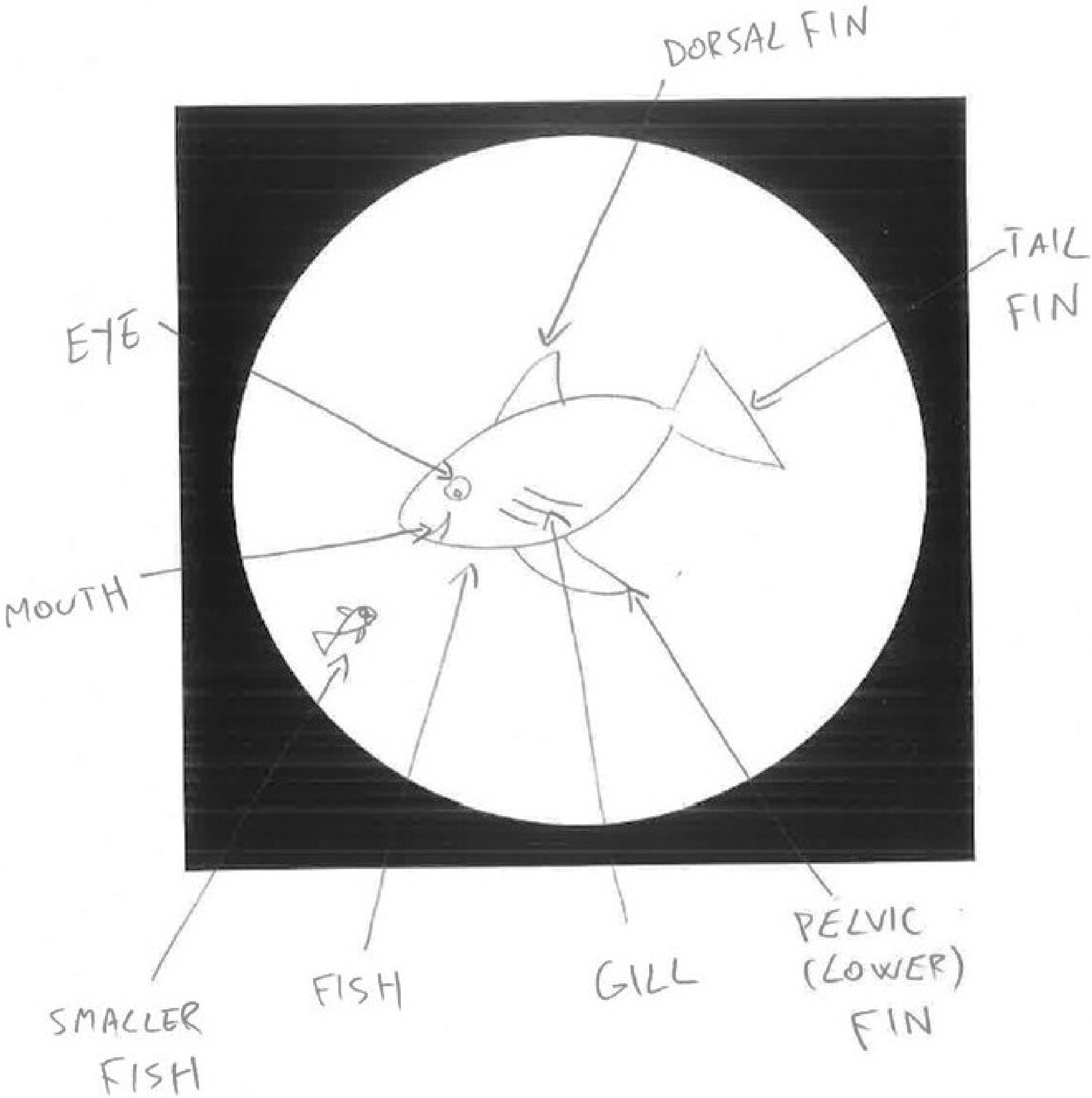
C	If we keep canned food in the refrigerator, it will be good for a prolonged period of time.		
D	If we keep canned food outside the refrigerator, even in the summer, it will be good for a prolonged period of time.	**X**	
E	Obviously closed bag of chips can stay outside the refrigerator for months, and will be good to eat. But, if we open the bag of chips, eat from it and then keep it at our desk for few days, the chips is still good to eat.		
F	Similarly, if you open a jar of honey, take a spoonful and eat it, then use the same spoon to take another, and finally a third taste using the same spoon, you would have transferred many bacteria from your mouth into the honey. However, even after two weeks, the honey will not spoil. Even during the summer and outside the refrigerator.		

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at both A, B and D, and nothing else. Anything else is not considered as correct answer, and students get 0 points.

STEP 3.5.

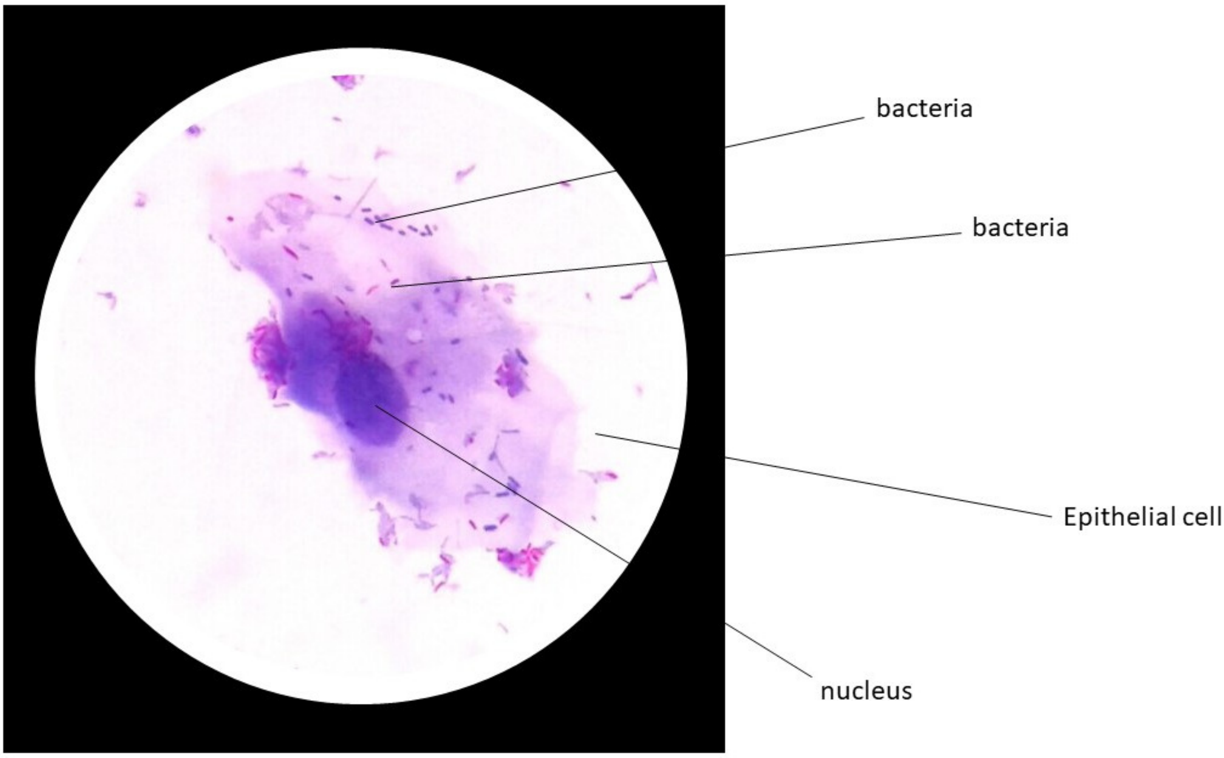
3.5.1. Using the immersion objective, carefully examine your stained slide and draw what you observe. Draw a representative image *within the circle (the field of view under the microscope). Be sure to include the important features, focusing on a field of view where both epithelial cells and bacterial cells are visible.*
Label the key features on your drawing. *Use arrows in your drawing to point to the important structures you should observe.* (5.5 p)

Example:
Below is an example of how your drawing might look.



For giving points to field view drawing we will do as following: We tested all slides, and certain key features will be visible, and students should be able to identify them (tested several times with our students). regardless of the language, we can see if they identified the key features by the drawings and arrows. Below are microscope images they will see (live microscope image is much nicer then this photography).

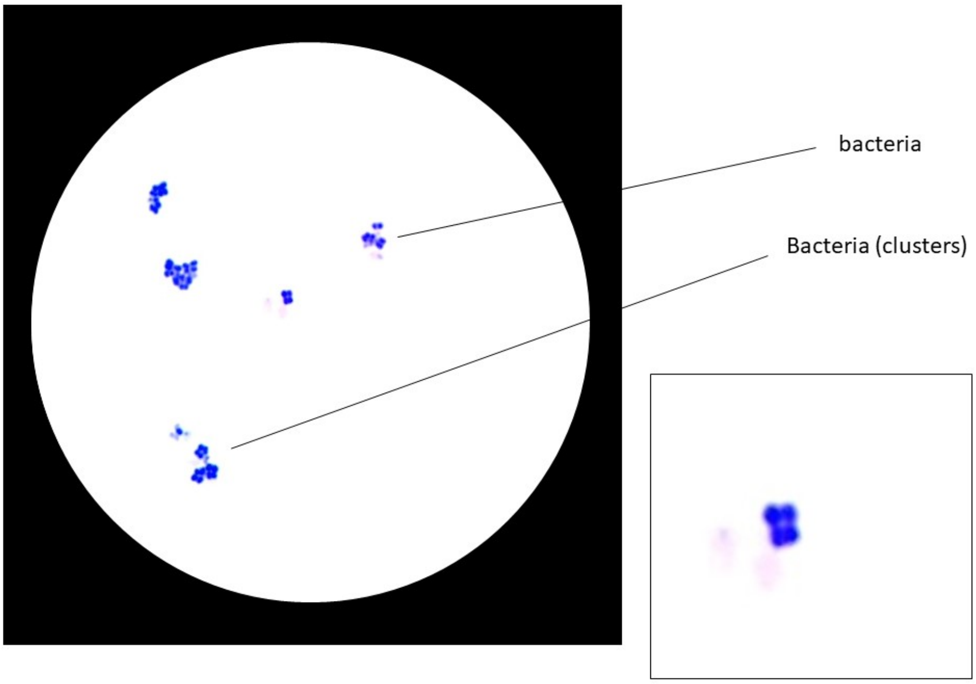
3.5.1. Representative image of your swab sample



This Table is to be filled by the reviewers	Marks
Student found the image (if not, Lab assistant must enter 0 here)	**1.5**
Image is drawn correctly	**1**
Key features	
Bacteria	**1**
Epithelial cell	**1**
Nucleus	**1**

STEP 3.6.

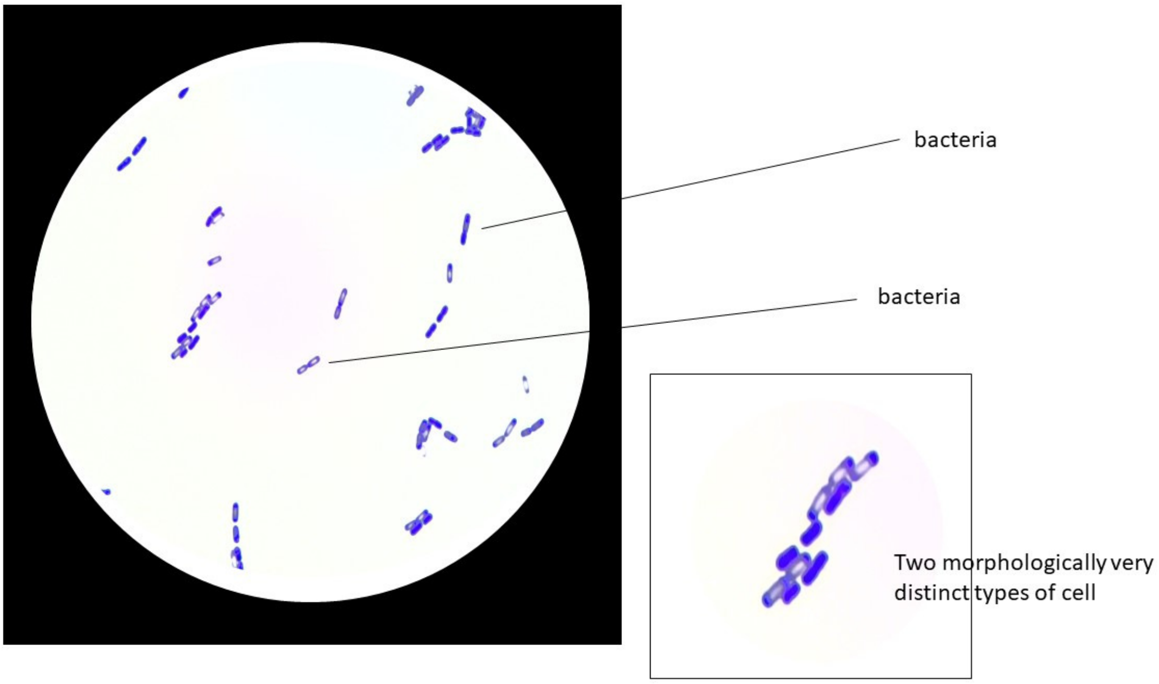
3.6.1. *Draw representative image of slide* ***marked X1****. Mark key features, if there are any.*
HINT *– Some bacteria divide in a way that their daughter cells remain attached, forming distinctive clusters with a constant geometric pattern, where the number of bacteria is always the same. Can you identify such clusters in your sample? If so, draw an enlarged schematic image in the square beneath the field of view.* (3 p)



Cluster of four cells, which is very specific for *Micrococcus luteus*. We will give point if students draw two cells as well, as sometimes this can be seen.

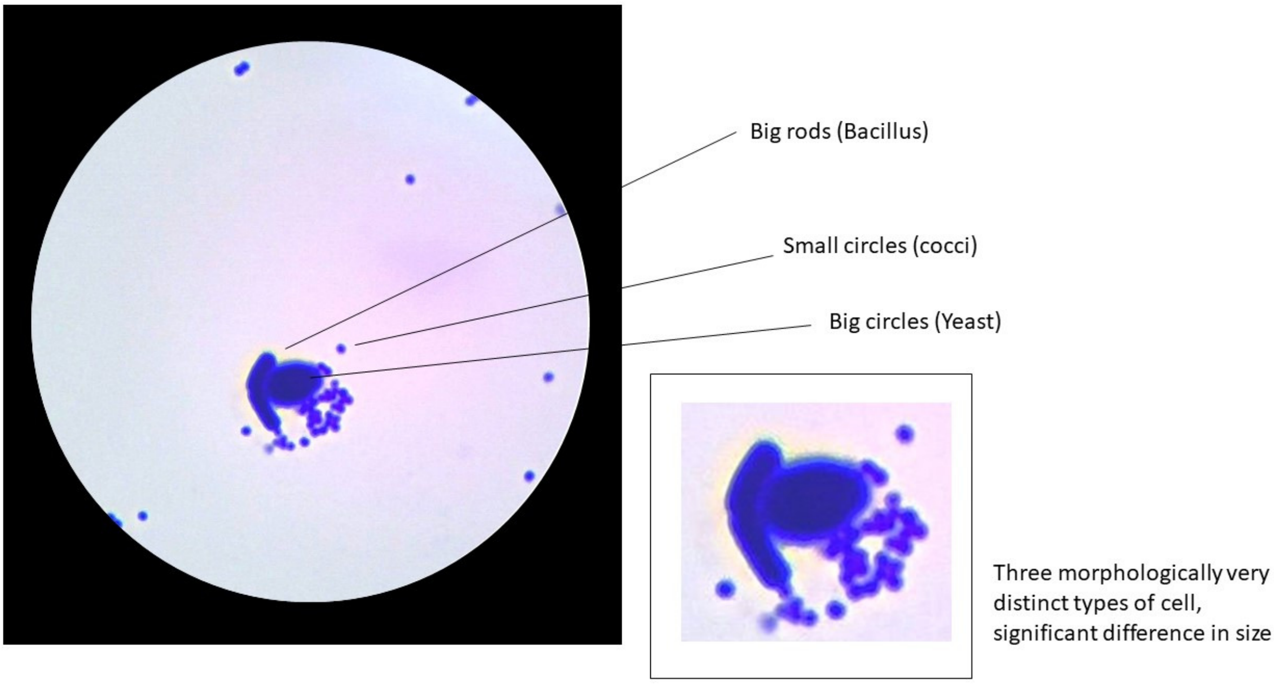
This Table is to be filled by the reviewers	**Marks**
Student found the image (if not, Lab assistant must enter 0 here)	**1**
Image is drawn correctly	**1**
Key features	
Tetrade (if drawn in the square)	**1**

3.6.2. *Draw representative image of slide* ***marked X2****. Mark key features, if there are any.*
HINT *– All the cells on this slide are of a single bacterial species, Bacillus cereus. But, although its the same species, there are two morphologically distinctive types of cells that you can see, if you look carefully. Can you find them and draw an enlarged schematic image in the square beneath the field of view?* (4 p)



This Table is to be filled by the reviewers	Marks
Student found the image (if not, Lab assistant must enter 0 here)	**1**
Image is drawn correctly	**1**
Key features	
Cell without spore (if drawn in the box)	**1**
Cell with spore (if drawn in the box)	**1**

3.6.3. *Draw representative image of slide* ***marked X3****. Mark key features, if there are any.*
HINT *– We prepared a mix of different microorganisms on this slide. If you can identify morphologically different cells, please do draw an enlarged schematic image of them in the square beneath the field of view? While drawing, try to keep the cell size ratio the same as you see it in the field of view.* (5 p)



This Table is to be filled by the reviewers	Marks
Student found the image (if not, Lab assistant must enter 0 here)	**1**
Image is drawn correctly	**1**
Key features	
Cocci (if drawn in the box)	**1**
Rods (if drawn in the box)	**1**
Round yeasts (if drawn in the box)	**1**

STEP 3.7.

3.7.1. Question: Take a good look at the image drawn at 3.5.1. If you know that nucleus is around 10 µm in diameter, could you estimate the average size of your bacterial cells? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
Around 10 nm		**1**
Around 100 nm		
Around 1 µm	**X**	
Around 0.01 mm		
Around 0.001 mm	**X**	

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at both Around 1 µm and around 0.001 mm, and nothing else. Anything else is not considered as correct answer, and students get 0 points.

3.7.2. Question: Observing the slide **X2** you should have been able to see two types of morphologically different bacterial cells. What do you think the difference was? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
They are from two different bacterial species.		**1**
Some cells have air bubbles inside them.		
Some cells have a feeding vacuole inside them.		
Some cells have enlarged nucleus due to division in progress.		
Some cells have endospores inside them.	**X**	

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at Some cells have endospores, and nothing else. Anything else is not considered as correct answer, and students get 0 points.

3.7.3. Question: Observing the slide **X3** you should have been able to see several types of morphologically different cells. What do you think what species they were? Mark the correct answer(s) with an X. (1 p)

Answer		Marks
Bacterium *Staphylococcus aureus*, bacterium *Olympiadus zagrabiensis* and yeast *Saccharomyces cerevisiae.*		**1**
Bacterium *Staphylococcus aureus*, bacterium *Bacillus megaterium* and protozoan *Paramecium caudatum.*		
Bacterium *Staphylococcus aureus*, bacterium *Staphylococcus argentum* and bacterium *Staphylococcus cuprum.*		
Bacterium *Staphylococcus aureus*, bacterium *Bacillus megaterium* and yeast *Saccharomyces cerevisiae.*	**X**	
Bacterium *Bacillus cocoideus*, bacterium *Bacillus megaterium* and protozoan *Euglena viridis.*		
Yeast *Saccharomyces cerevisiae*, yeast *Saccharomyces cocoidensis* and yeast *Saccharomyces bacillae.*		

If the answer is correct student will get full point (1.0 point), if it is wrong students gets 0 point. Answer is correct only if they marked X at place marked above, and nothing else. Anything else is not considered as correct answer, and students get 0 points. in this question students do not need to be familiar with any of the listed species. They can give correct answer just by elimination.